Reader's Digest and Regurgitation: A Multi-Dimensional NLP Pipeline for News Genre Classification, Keyword Extraction, and Sentiment Analysis

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Abstract

*We present **NewsScope**, a multi-task NLP pipeline for automatic analysis of BBC News articles. The system addresses three complementary problems: genre classification, keyword and named-entity extraction, and sentiment analysis. Genre classification is performed by a custom Multinomial Naive Bayes classifier trained on the BBC News corpus [2], achieving 98.24% accuracy across five categories (business, entertainment, politics, sport, and technology) on a held-out test set. Keyword extraction combines TF-IDF ranking with part-of-speech filtering and NLTK's named-entity chunker [1]. Sentiment analysis uses a second Naive Bayes model trained on BBC articles pseudo-labeled by the VADER rule-based analyzer [3], keeping the sentiment model in-domain with the news register. The full pipeline is demonstrated live on a BBC RSS feed. Our results show that bag-of-words Naive Bayes is a surprisingly strong baseline for news genre classification, while sentiment analysis in the news domain remains a harder problem due to its predominantly neutral, factual tone.*

1. Introduction

The volume of news produced online every day is staggering. Readers, researchers, and media organizations alike need ways to quickly understand what a given article is about and whether its tone is positive or negative. Doing this manually at scale is simply not feasible, which makes automated news analysis a practically important problem and a rich domain for applying NLP techniques.

Genre classification — assigning a topic category such as *sport* or *technology* to an article — is the most fundamental of these tasks. It underpins content recommendation systems, archive organization, and media monitoring. Beyond genre, two additional layers of information are particularly useful: (1) the specific terms and entities that distinguish an article from the background corpus, and (2) the overall sentiment of the coverage, which can reveal editorial tone or public mood around a topic.

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This project builds an end-to-end pipeline, which we call **NewsScope**, that addresses all three tasks in sequence:

1. **Genre classification:** A Multinomial Naive Bayes classifier assigns each article to one of five BBC categories.

2. **Keyword and entity extraction:** TF-IDF scoring, POS-tag filtering, and named-entity recognition surface the most important terms and real-world entities in each article.

3. **Sentiment analysis:** A Naive Bayes sentiment model trained on in-domain BBC data (pseudo-labeled by VADER) is compared against VADER's rule-based scores directly.

The entire pipeline is implemented using NLTK [1] and scikit-learn, and demonstrated on live BBC RSS headlines.

The rest of this report is organized as follows. Section 3 describes the dataset. Section 4 details the preprocessing, feature extraction, and modeling choices. Section 5 reports quantitative results and example outputs. Section 6 discusses the findings and limitations. Section 7 concludes.

2. Related Work

Text classification has a long history in NLP. Naive Bayes applied to bag-of-words representations was popularized by McCallum and Nigam [4], who showed that even with its independence assumption, it achieves competitive results on document categorization benchmarks. Despite the rise of deep learning, it remains a strong baseline for topic classification due to its efficiency and interpretability — properties that make it well-suited for an educational NLP project.

TF-IDF as a term weighting scheme for information retrieval is a well-established technique in the field [1]. For keyword extraction, it complements POS-tag filtering naturally: TF-IDF handles corpus-level rarity while POS tags restrict terms to content-bearing word classes (nouns and adjectives), together yielding more human-readable keywords than either method alone.

Sentiment analysis of news text is notably harder than sentiment analysis of reviews or social media, because news writing is deliberately neutral and factual. Hutto and Gilbert [3] introduced VADER (Valence Aware Dictionary and sEntiment Reasoner), a lexicon and rule-based method tuned on social media text that generalizes to other domains including news headlines. Using VADER to pseudo-label a training corpus is an established approach for building lightweight in-domain sentiment classifiers when gold-standard labels are unavailable.

3. Data

3.1. BBC News Corpus

We use the BBC News dataset introduced by Greene and Cunningham [2], a widely-used benchmark for topic classification. The corpus contains 2,225 news articles collected from the BBC website between 2004 and 2005, spanning five editorial categories: *business*, *entertainment*, *politics*, *sport*, and *tech*. The category distribution is shown in Table 1.

Category	**Articles**	**Share (%)**
Business	510	22.9
Sport	511	23.0
Tech	401	18.0
Politics	417	18.7
Entertainment	386	17.3
Total	**2,225**	**100.0**

Table 1. BBC News dataset category distribution. Business and sport are the largest categories; entertainment is the smallest.

The articles are relatively short (median length approximately 300 words), topically diverse within each category, and written in a formal news register. Because BBC articles are carefully edited, the corpus has relatively few typos or non-standard forms, which simplifies preprocessing. The dataset is included with the project as `bbc.zip` and is loaded from the extracted `bbc/` directory.

3.2. Train / Validation / Test Split

We apply a stratified split to preserve the class distribution in every partition. The dataset is divided into 80% training, 10% validation, and 10% test, yielding approximately 1,777 training, 221 validation, and 227 test articles. Stratification ensures that even the smaller categories (entertainment, tech) are adequately represented in every partition. The random seed is fixed at 42 for reproducibility.

3.3. Sentiment Labels

No gold-standard sentiment labels exist for the BBC corpus. Instead, we generate pseudo-labels using VADER [3]: articles with a compound score ≥ 0.05 are labeled *positive*, those ≤ -0.05 are labeled *negative*, and strictly neutral articles are discarded. This produces a binary sentiment dataset drawn entirely from BBC text, keeping the vocabulary of the sentiment model consistent with the genre classifier's training domain.

4. Method

4.1. Preprocessing

All articles pass through a shared preprocessing step before any modeling. Each article is split on whitespace, then each token is:

1. lowercased,
2. filtered to retain only tokens matching the pattern `[^a-zA-Z0-9]` (i.e., alphabetic-initial tokens),
3. filtered to remove tokens shorter than 2 characters,
4. optionally filtered against an English stopword list (NLTK `stopwords.words("english")`).

Stopword removal is applied for genre classification and keyword scoring, where function words carry no topical signal. It is *not* applied for sentiment classification, where negation words ("not," "never") carry meaningful polarity information.

Unlike many production NLP pipelines, we deliberately avoid stemming or lemmatization. Retaining full word forms keeps the vocabulary human-readable in keyword output and avoids the risk of conflating morphologically similar but semantically distinct terms (e.g., *bank* and *banking* in a business context).

4.2. Feature Extraction

Bag-of-Words for classification. Both the genre and sentiment classifiers represent documents as bags of words — unordered multisets of tokens after preprocessing. When `binarize=True` (disabled by default), each token type is counted at most once per document, converting the representation to a binary presence/absence feature vector. In practice we found standard token counts to perform comparably to binarized counts on this corpus.

TF-IDF for keyword ranking. For keyword extraction we build an inverted index over all 2,225 articles and compute TF-IDF weights using the standard log-frequency formulation:

$$\text{tf-idf}(t,d) = (1 + \log_{10} \text{tf}_{t,d}) \times \log_{10} \frac{N}{\text{df}_t}$$

where $\text{tf}_{t,d}$ is the raw term frequency in document d, N is the total number of documents, and df_t is the number of documents containing term t. For articles not seen during index construction (e.g., live RSS articles), we reuse the corpus IDF values and compute TF-IDF on the fly.

POS-tag filtering. To produce more readable keyword lists, we additionally filter terms by their NLTK part-of-speech tag, retaining only nouns (NN, NNS, NNP, NNPS) and adjectives (JJ, JJR, JJS). This removes verbs, determiners, and other grammatical function words that might receive high TF-IDF scores in short snippets.

4.3. Named Entity Recognition

Named entities are extracted using NLTK's `ne_chunk` function [1], which applies a binary Maximum Entropy NE chunker on top of POS-tagged tokens. We extract entities of types PERSON, ORGANIZATION, GPE (geopolitical entity), GSP (geographical/social/political), and FACILITY. Each article's entities are de-duplicated and grouped by type before display.

4.4. Genre Classifier

We implement Multinomial Naive Bayes from scratch following the derivation in McCallum and Nigam [4]. For a document d with token list $\mathbf{w}$ and class set $\mathcal{C}$, the predicted class is:

$$\hat{c} = \arg\max_{c \in \mathcal{C}} \left[\log P(c) + \sum_{w \in \mathbf{w}} \log P(w \mid c) \right]$$

Parameters $P(c)$ are estimated by relative class frequency in the training set. Conditional probabilities $P(w \mid c)$ are estimated with Laplace (add-α) smoothing:

$$P(w \mid c) = \frac{\text{count}(w, c) + \alpha}{\sum_{w'} \text{count}(w', c) + \alpha \cdot |V|}$$

where $|V|$ is the vocabulary size and $\alpha = 1.0$ by default. All computations are done in log-space to avoid floating-point underflow. At inference time, words outside the training vocabulary are silently skipped (their log-probability contribution is zero).

4.5. Sentiment Analysis

Sentiment classification uses two independent systems whose outputs are compared:

- **VADER** [3]: a rule-based analyzer using a curated sentiment lexicon with heuristics for capitalization, punctuation, and valence shifters (negation, degree adverbs). It produces a compound score in $[-1, +1]$; we threshold at ± 0.05 to assign positive/negative/neutral labels.

- **NB Sentiment**: the same Multinomial Naive Bayes architecture used for genre classification, trained on BBC articles pseudo-labeled by VADER (neutral articles excluded). Training on in-domain text means the model has seen the vocabulary and syntactic patterns of news writing, which we hypothesize should produce more reliable sentiment predictions than a model trained on movie reviews or social media.

4.6. Why NLTK?

NLTK [1] was chosen as the primary NLP toolkit for several reasons. First, it provides all the components needed in a single, well-documented library: tokenization, POS tagging, named-entity chunking, the VADER sentiment analyzer, and access to standard corpora. Second, its APIs closely match the conceptual descriptions in the *NLTK Book* [1], making the implementation transparent and educational. Third, its rule-based taggers and chunkers are deterministic and require no additional trained model downloads beyond the standard data packages, making the project straightforward to set up. Where statistical learning is required (the genre and sentiment classifiers), we use our own implementation rather than wrapping an NLTK class, to demonstrate that the core algorithm is understandable and not a black box.

5. Experiments

5.1. Genre Classification

Cross-validation. We evaluate both classifiers using stratified 5-fold cross-validation on the full corpus before committing to a final model. Each fold preserves the original class distribution. Table 2 summarizes the per-fold accuracy for both the genre and sentiment classifiers.

Fold	Genre Acc.	Sentiment Acc.
Fold 1	0.9685	0.7760
Fold 2	0.9820	0.8009
Fold 3	0.9730	0.7964
Fold 4	0.9730	0.8118
Fold 5	0.9775	0.7959
Mean	**0.9748**	**0.7962**

Table 2. Stratified 5-fold cross-validation accuracy for both classifiers. Sentiment CV uses BBC articles pseudo-labeled by VADER.

Test set results. The final model is trained on the full training partition (80%, ≈1,777 articles) and evaluated on the held-out test set (10%, 227 articles). It achieves **98.24% accuracy**. Per-class precision, recall, and F1 are shown in Table 3.

Category	Prec.	Recall	F1	Support
Business	1.0000	0.9608	0.9800	51
Entertainment	1.0000	0.9750	0.9873	40
Politics	0.9556	1.0000	0.9773	43
Sport	0.9811	1.0000	0.9905	52
Tech	0.9756	0.9756	0.9756	41
Weighted avg	**0.9829**	**0.9824**	**0.9824**	**227**

Table 3. Per-class classification report on the 227-article test set.

Confusion matrix. The confusion matrix (Table 4) reveals that the vast majority of errors are between adjacent topic areas — a business article misclassified as politics, or a sport article misclassified as tech. There are no gross misclassifications (e.g., sport predicted as entertainment).

True \ Pred	Bus.	Ent.	Pol.	Spt.	Tch.
Business	**49**	0	1	0	1
Entertainment	0	**39**	1	0	0
Politics	0	0	**43**	0	0
Sport	0	0	0	**52**	0
Tech	0	0	0	1	**40**

Table 4. Confusion matrix on the test set. Rows = true labels, columns = predicted. Diagonal entries (bold) are correct predictions.

5.2. Keyword and Entity Extraction

We present three sample articles from the corpus to illustrate the keyword and NER pipeline. Each article's top-8 TF-IDF keywords, top-8 POS-filtered keywords, and named entities are shown in Table 5.

Several observations stand out. TF-IDF keywords tend to capture proper nouns and rare compound forms (e.g., *yearearlier*, *moscowbased*) that are unique to the article, while POS-filtered keywords tend to be more general and readable. The NER output correctly identifies organizations and people but occasionally misclassifies common nouns as geopolitical entities (e.g., "Ad" as a GPE), a known limitation of NLTK's rule-based chunker.

5.3. Sentiment Analysis

Table 6 shows the sentiment predictions for the same three articles from both the NB model and VADER.

Agreement between the two systems is mixed. On the three samples, two agree and one disagrees — a pattern we observe broadly across the corpus. We examine the causes of disagreement in Section 6.

6. Discussion

6.1. Which Genres Are Easiest and Hardest?

Looking at the confusion matrix (Table 4), sport and politics achieve perfect recall (1.0000), meaning every sport and politics article in the test set was correctly identified. This is intuitive: sport articles have a very distinctive vocabulary — the names of specific athletes, teams, scores, and sport-specific terminology (e.g., *innings*, *championship*, *goalscorer*) rarely appear in other categories. Politics similarly revolves around a narrow lexicon of government institutions, party names, and legislative concepts.

Business and tech are the hardest pair. Two business articles were misclassified: one as politics (likely a story about economic policy or government spending) and one as tech (probably a story about a technology company's earnings). This reflects genuine topical overlap — a story about, say, Google's stock performance or a government stimulus package spans multiple editorial categories by nature. Entertainment had one misclassification as politics, possibly a story about a celebrity's political activities.

Overall, these errors suggest that the biggest limitation of the bag-of-words approach is its insensitivity to context: a word like *revenue* appears in both business and tech articles, and without syntactic or semantic context the model cannot distinguish them.

6.2. Sentiment Limitations in the News Domain

News articles present a fundamentally different challenge for sentiment analysis compared to the domains (product reviews, movie reviews, social media) on which most sentiment tools are developed. News writing is deliberately neutral: journalists are trained to present facts without editorializing, which means the majority of BBC articles land close to sentiment-neutral regardless of their topic.

This creates two practical problems. First, VADER's compound scores for many articles cluster around zero, producing many *neutral* labels that we discard during training — reducing the size of the sentiment training set and potentially biasing the NB model toward the articles that are most polarized. Second, when NB and VADER disagree, it is often because the article contains factual language that VADER's lexicon parses as positive (e.g., words like *profit*, *growth*, *rise*) while the NB model, trained on BBC text, has learned that these words appear across both positive and negative business stories.

Article	TF-IDF Keywords	POS Keywords	Named Entities
Time Warner profits (business)	timewarner, yearearlier, aols, aol, restate, fullyear, 639m, 111bn	timewarner, aol, profits, sales, warner, internet, profit, time	GPE: Ad, Google; PERSON: Time Warner, Warner; ORG: TimeWarner
Dollar/Greenspan (business)	greenspan, 12871, 12974, greenspans, sinche, loosening, halfpoint, deficit	dollar, deficit, months, greenspan, federal, reserve, current	PERSON: Alan Greenspan; ORG: Federal Reserve; FACILITY: White House
Yukos/Rosneft (business)	yugansk, rosneft, menatep, yukos, 479m, 540m, rosnefts, moscow-based	rosneft, yukos, sale, unit, loan, menatep, assets	GPE: Russian; ORG: Yukos, Menatep Group

Table 5. Sample keyword and entity extraction output for three business articles from the corpus.

Article	NB	VADER	Compound
Time Warner profits	pos	pos	+0.9954
Dollar/Greenspan	neg	pos	+0.9421
Yukos/Rosneft	neg	neg	−0.8860

Table 6. NB and VADER sentiment predictions with VADER compound scores for three sample articles. The second article shows a NB/VADER disagreement.

6.3. Sarcasm, Ambiguity, and Bias

Both the NB model and VADER fail in systematic ways that are worth acknowledging:

Sarcasm and irony. Neither model has any representation of sarcasm. A headline like "Government's brilliant plan to fix economy" would be scored positively by VADER despite potentially being ironic. Detecting sarcasm typically requires pragmatic context well beyond what bag-of-words or lexicon-based methods can capture.

Negation scope. VADER applies simple heuristics for negation (e.g., flipping polarity when preceded by "not"), but complex negation (*"not entirely unsuccessful"*) is frequently mishandled. The NB model ignores negation entirely, since tokens are treated as independent features.

Domain bias in pseudo-labels. Because the NB sentiment model's training labels come from VADER, any systematic errors in VADER's lexicon are inherited by the NB model. If VADER tends to score financial vocabulary as positive (due to words like *gain*, *boost*, *profit*), the NB model will learn that business vocabulary is positive — even when the article is reporting a corporate scandal.

Representation bias. The BBC corpus reflects the editorial stance and topic selection of a single British public broadcaster. The sentiment model trained on this corpus cannot be assumed to generalize to news from other outlets, countries, or political leanings.

6.4. NER Quality

The named-entity extractor performs reasonably well on proper nouns but has two recurring weaknesses. First, it occasionally misclassifies common nouns or short words as geopolitical entities (the "Ad" GPE example in Table 5). Second, multi-token entity boundaries are sometimes incorrect — for instance, splitting "Alan Greenspan" into two separate entities or merging adjacent entities. These are known limitations of the NLTK rule-based chunker, which uses context-free patterns rather than a sequence model trained on large annotated corpora.

7. Conclusion

We presented NewsScope, a multi-task NLP pipeline for automatic BBC news article analysis. The system demonstrates that a custom Multinomial Naive Bayes classifier, trained on bag-of-words features with Laplace smoothing, achieves near-ceiling accuracy (98.24%) on the BBC genre classification task. The pipeline additionally extracts meaningful keywords via TF-IDF and POS filtering, identifies named entities using NLTK's chunker, and produces sentiment predictions using both a rule-based analyzer (VADER) and an in-domain Naive Bayes model trained on BBC pseudo-labels.

What worked. Genre classification is the clear success of this project. The results confirm that BBC news categories have distinct enough vocabularies that a simple bag-of-words model generalizes almost perfectly to unseen articles. The combination of TF-IDF keywords and POS-tag filtering produced readable and relevant keyword lists, and the named entity extractor successfully identified the key people, organizations, and places discussed in most articles. Keeping the sentiment NB model in-domain (trained on BBC text rather than movie reviews) was the right design choice: the vocabulary overlap with the genre classifier ensures that the model is not penalized for news-specific phrasing.

What could be improved.

- **Sentiment quality.** The biggest limitation is sentiment analysis. News text is overwhelmingly neutral, and pseudo-labels derived from VADER are noisy. A small amount of human-annotated sentiment data — even a few hundred articles — would likely improve performance substantially.
- **Feature enrichment.** Beyond unigrams, bigram features would capture phrases like *interest rate* or *match result* that carry topical meaning lost in the bag-of-words representation.
- **NER accuracy.** Replacing the rule-based NLTK chunker with a modern sequence-labeling model would significantly reduce the boundary errors and false-positive entity detections observed in our outputs.

Overall, this project demonstrates that classical NLP methods — properly implemented and applied to a well-matched dataset — can achieve strong results on genre classification while also providing interpretable outputs (keywords, entities, sentiment) that would be difficult to extract from a black-box neural model. The pipeline is compact, self-contained, and fully reproducible from the included `bbc.zip` dataset.

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